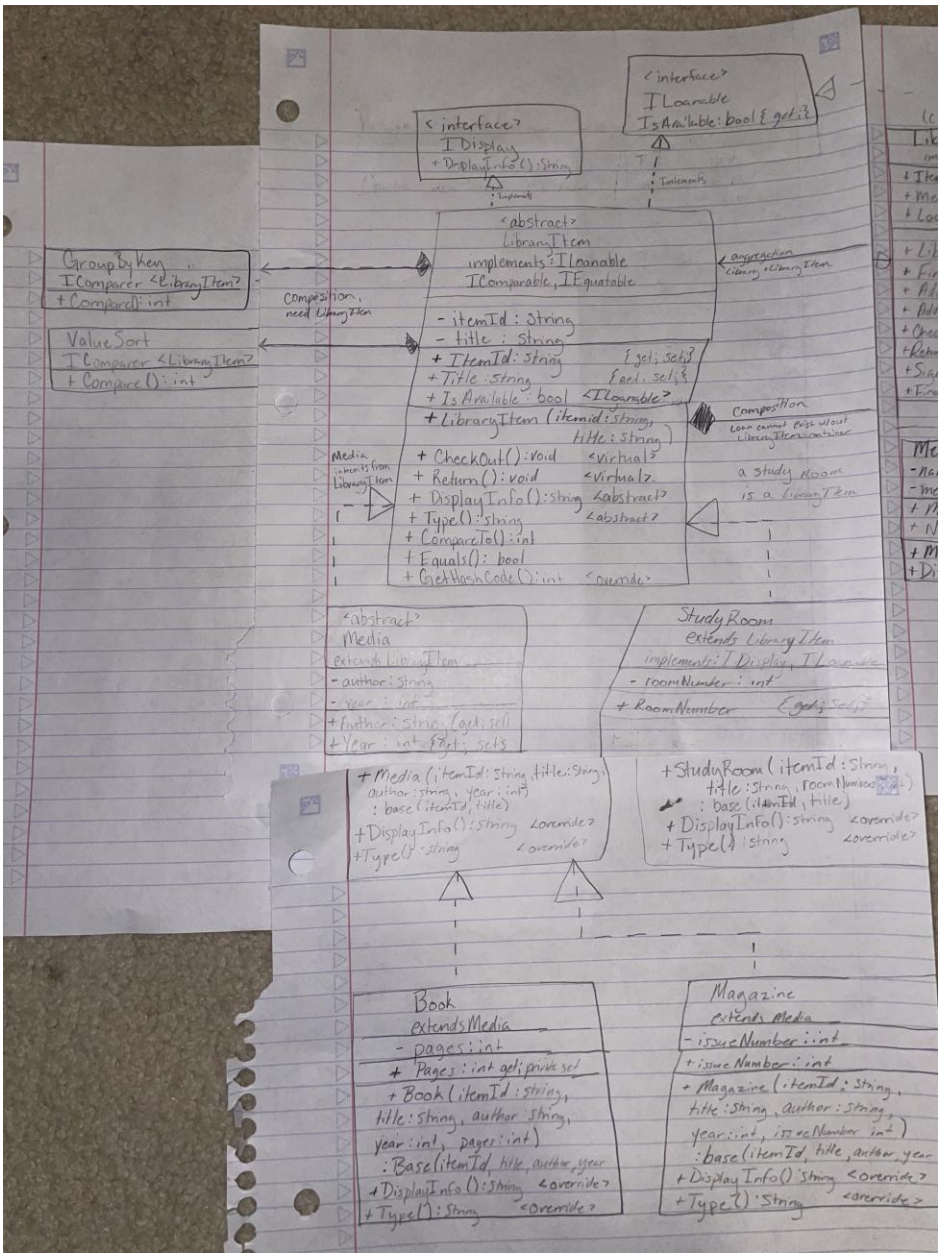
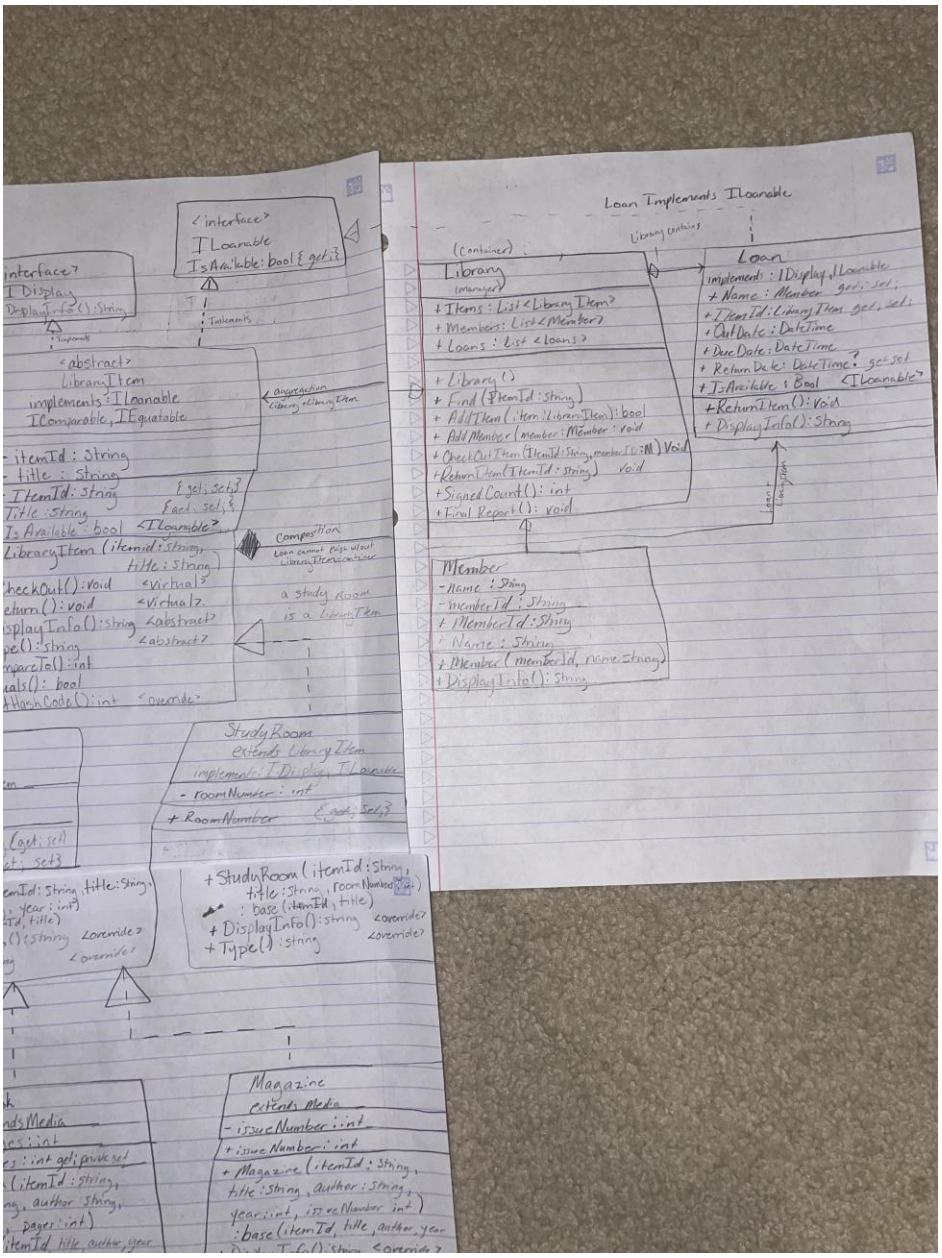

<interface>
IDisplay
+DisplayInfo():string
<interface>
ILoanable
IsAvailable: bool {get;}
Implements
Implements
<abstract>
LibraryItem
implements: ILoanable
IComparable, IEquatable
- itemId : String
- title : String
+ ItemId: string {get; set;}
+ Title : string {get; set;}
+ IsAvailable : bool <ILoanable>
+ LibraryItem (itemid: string, title: string)
+ CheckOut(): void <virtual>
+ Return(): void <virtual>
+ DisplayInfo(): string <abstract>
+ Type(): string <abstract>
+ CompareTo(): int
+ Equals(): bool
+ GetHashCode(): int <override>
GroupByKey
IComparer <LibraryItem>
+Compare(): int
ValueSort
IComparer <LibraryItem>
+ Compare(): int
Composition,
need LibraryItem
Aggregation
Library + LibraryItem
Composition
Loan cannot exist without
LibraryItem in Loan
a Study Room
is a LibraryItem
Media
...inherits from
LibraryItem
<abstract>
Media
extends LibraryItem
- author: string
- year : int
+ Author : string {get; set;}
+ Year : int {get; sets}
Study Room
extends LibraryItem
implements: IDisplay, ILoanable
- roomNumber : int
+ RoomNumber {get; set;}
+ Media (itemId: string, title: string,
author: string, year: int)
: base (itemId, title)
+ DisplayInfo(): string <override>
+ Type() :string <override>
+ StudyRoom (itemId: string,
title: string, roomNumber: int)
: base (itemId, title)
+ DisplayInfo(): string <override>
+ Type() :string <override>
Book
extends Media
- pages : int
+ Pages : int {get; private set}
+ Book (itemId : string,
title: string, author: string,
year: int, pages: int)
: base (itemId, title, author, year
+ DisplayInfo(): string <override>
+ Type(): string <override>
Magazine
extends Media
- issueNumber : int
+ issueNumber : int
+ Magazine (itemId: string,
title: string, author: string,
year: int, issueNumber: int)
: base (itemId, title, author, year
+ DisplayInfo() string <override>
+ Type() : string <override>


Loan Implements ILoanable
<interface>
ILoanable
IsAvailable: bool { get; }
Implements
<abstract>
LibraryItem
implements: ILoanable
IComparable, IEquatable
itemId : String
title : String
ItemId: String { get; set; }
Title : String { get; set; }
IsAvailable: bool
LibraryItem (itemId: String, title: String)
CheckOut(): void <virtual>
Return(): void <virtual>
DisplayInfo(): String <abstract>
Type(): String <abstract>
CompareTo(): int
Equals(): bool
GetHashCode(): int <override>
Aggregation
Library contains LibraryItem
Composition
Loan cannot exist without LibraryItem/container
a Study Room is a LibraryItem
(container)
Library
(managed)
+ Items : List<LibraryItem>
+ Members: List<Member>
+ Loans : List<Loans>
+ Library()
+ Find(ItemId : String)
+ AddItem(item: LibraryItem): bool
+ AddMember(member: Member): void
+ CheckOutItem(ItemId: String, memberID: M) Void
+ ReturnItem(ItemId: String) void
+ SignedCount(): int
+ FinalReport(): void
Library contains
Loan
implements: IDisplay, ILoanable
+ Name : Member get; set;
+ ItemId: LibraryItem get; set;
+ OutDate : DateTime
+ DueDate : DateTime
+ ReturnDate: DateTime get; set;
+ IsAvailable : Bool <ILoanable>
+ ReturnItem(): Void
+ DisplayInfo(): String
Loan is Loaning
Member
- Name : String
- MemberId : String
+ MemberId: String
+ Name : String
+ Member(memberId, Name String)
+ DisplayInfo(): String
StudyRoom
extends LibraryItem
implements: IDisplay, ILoanable
- RoomNumber : int
+ RoomNumber { get; set; }
+ StudyRoom(itemId: String, title: String, roomNumber: int)
: base(itemId, title)
+ DisplayInfo(): string <override>
+ Type(): string <override>
Magazine
extends Media
- issueNumber : int
+ issueNumber : int
+ Magazine(itemId: String, title: String, author: String, year: int, issueNumber: int)
: base(itemId, title, author, year